11.a.) Simulate static Routing using CISCO Packet Tracker.

24/9/26

Aim:

Simulate static Routing configuration using CISCO

Theory:

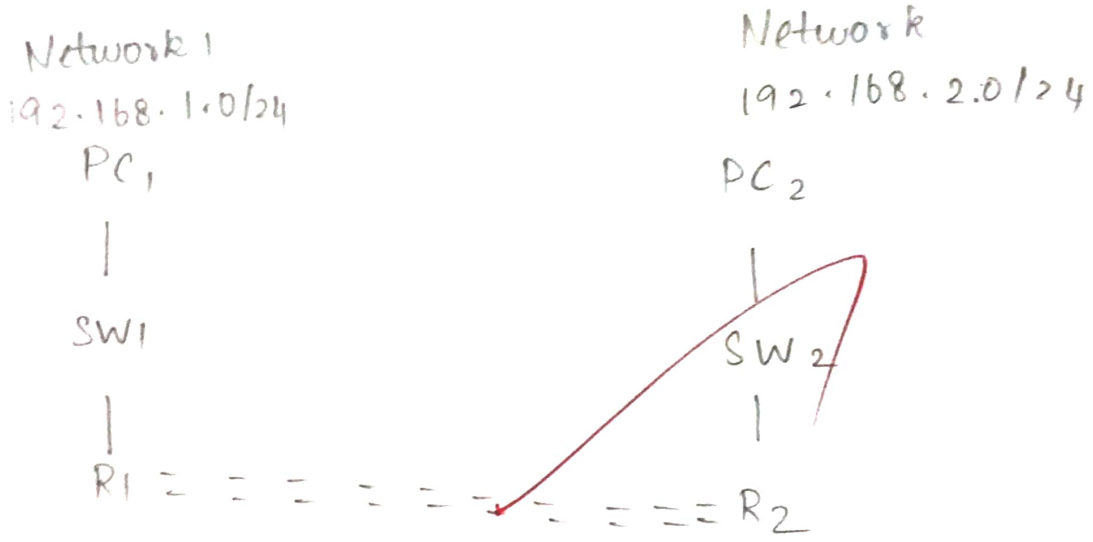
Static routing is the process of manually adding routes to a routers's routing table. The router automatically knows only the network directly connected to its interfaces. Routes to other networks must be configured manually.

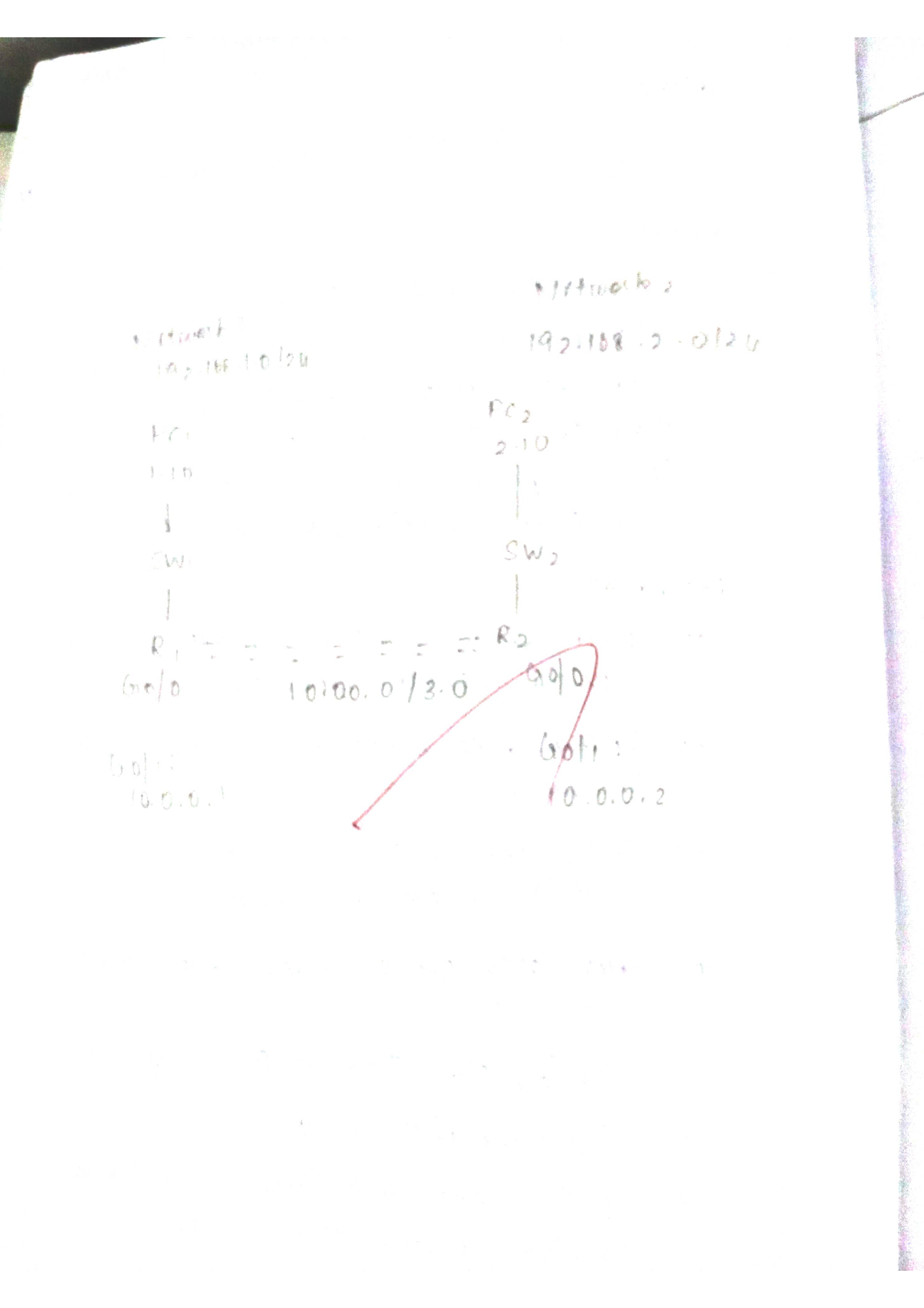
PROCEDURE

1. Open CISCO packet Tracer and create the required network topology.
2. Configure IP addresses for the router interfaces and PCs.
3. Enter the router CLI using enable configure terminal
4. Add static routes using the ip route command.
5. Verify the static routes using
6. Test connectivity using.
7. If the main route fails, the backup route is automatically used.

Result: Thus, static routing was successfully configured and verified using CISCO Packet Tracer.

Bag 2/9


Network 1
192.168.1.0/24
PC1
SW1
R1
Network
192.168.2.0/24
PC2
SW2
R2


PC2
2.10
SW2
R2
10.00.0/30
Gi0/0
Gi0/1:
10.0.0.2

11. b.

RIP routing

Aim: Simulate RIP routing using CISCO packet Tracer.

Theory:

RIP (Routing Information Protocol) is a dynamic routing protocol used to exchange routing information between routers

Procedure:

1. Create the required network topology in CISCO Packet Tracer.
2. Assign IP address to Pc and router interfaces.
3. Enable RIP using the router RIP Command
4. Configure RIP on all routers
5. Test connectivity between PC0 and PC1 using the Ping command.
6. If one route fails, RIP automatically selects another available route.

Command:

Router:

```
router rip
network 10.0.0.0
network 192.168.1.252
network 192.168.1.248
```

Result:

Thus, RIP routing was successfully configured and verified using CISCO packet Tracer.